

HW#5 Tianpu Zhao

#1

(1) please see the attached files for the RTL and test bench files (Q1.v and Q1_tb.v), the timing constraint file (.sdc) is not supported on canvas upload, so I paste here:

```
// Q1_sdc.sdc
```

```
create_clock -period 1.0 [get_ports clk]
```

```
set_input_delay -max 0.5 -clock clk [get_ports {IN W}]
```

```
set_input_delay -min -0.2 -clock clk [get_ports {IN W}]
```

```
set_output_delay -max 0.5 -clock clk [get_ports {OUT}]
```

```
set_output_delay -min -0.2 -clock clk [get_ports {OUT}]
```

(2) please see the screenshot in the following pages

(3) please see the screenshots in the following pages. Timing slack is 43 ps,

```

1  =====
2  Generated by:      Genus(TM) Synthesis Solution 18.14-s037_1
3  Generated on:      Nov 18 2025  08:52:38 pm
4  Module:           mac
5  Operating conditions:  typical
6  Interconnect mode:  global
7  Area mode:        physical library
8  =====
9
10 Instance Module   Cell Count   Cell Area   Net Area   Total Area
11 -----
12 mac               124         348.460    168.121    516.581

```

```

=====
1   Generated by:      Genus(TM) Synthesis Solution 18.14-s037_1
2   Generated on:      Nov 18 2025  08:52:26 pm
3   Module:           mac
4   Operating conditions: typical
5   Interconnect mode: global
6   Area mode:        physical library
7   =====
8
9
10
11 Path 1: MET (43 ps) Setup Check with Pin mult_reg_reg[6]/CK->D
12 |    Group: clk
13 | Startpoint: (R) W[1]
14 |    Clock: (R) clk
15 | Endpoint: (R) mult_reg_reg[6]/D
16 |    Clock: (R) clk
17 |
18 | Capture          Launch
19 | Clock Edge:+     1000         0
20 | Drv Adjust:+       0         0
21 | Src Latency:+     0         0
22 | Net Latency:+     0 (I)      0 (I)
23 | Arrival:=       1000         0
24 |
25 | Setup:-          35
26 | Required Time:=   965
27 | Launch Clock:-    0
28 | Input Delay:-     500
29 | Data Path:-       422
30 | Slack:=          43
31 |
32 Exceptions/Constraints:
33 | input_delay       500              Q1_sdc.sdc_line_3_6_1

```

```

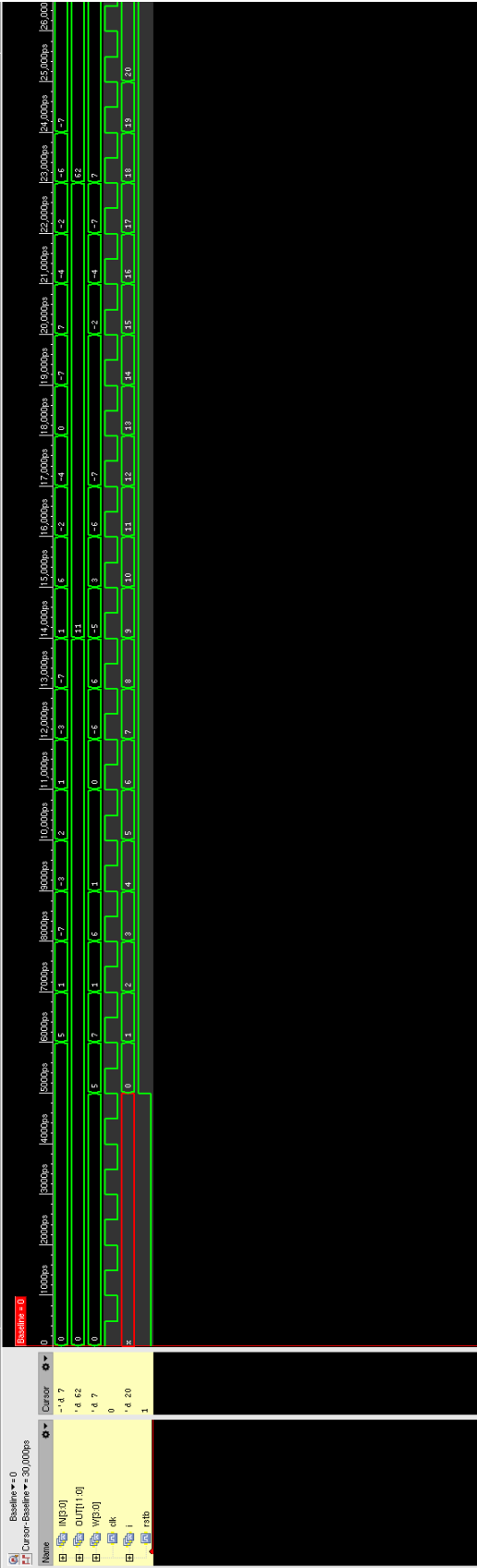
32 Exceptions/Constraints:
33 | input_delay          500          Q1_sdc.sdc_line_3_6_1
34
35 #-----
36 # Timing Point      Flags   Arc   Edge   Cell      Fanout Load Trans Delay
37 # Arrival Instance                                     (fF) (ps) (ps)
38 # (ps) Location
39 #-----
39 | W[1]              -      -      R      (arrival)    4  5.4    0
40 | 0          500      (-,-)
40 | g2324__7344/ZN    -      A2->ZN R      AND2_X1      1  3.5    12
40 | 33          533      (-,-)
41 | g2310__1474/S    -      A->S   R      HA_X1        1  3.1    28
41 | 55          588      (-,-)
42 | g2297__6083/S    -      CI->S  F      FA_X1        1  3.0    16
42 | 99          687      (-,-)
43 | g2280__2391/C0   -      CI->C0 F      FA_X1        1  3.0    15
43 | 73          760      (-,-)
44 | g2278__1309/C0   -      CI->C0 F      FA_X1        2  4.1    17
44 | 75          835      (-,-)
45 | g2274__2683/ZN   -      B1->ZN R      OAI21_X1     2  4.4    31
45 | 44          878      (-,-)
46 | g2268__3772/ZN   -      A->ZN  R      XNOR2_X1     1  1.4    17
46 | 43          922      (-,-)
47 | mult_reg_reg[6]/D <<< -      R      DFFR_X1      1  -      -
47 | 0          922      (-,-)
48 #-----

```

waveform (RTL):



waveform (synthesized):



#2

(a) $T_C = 5\text{ps}$. Limit on t_{setup} (max. possible value) is:

$$t_{\text{setup}} + t_{\text{pdmax}} + t_{\text{pcq}} = 5\text{ps}.$$

$$\begin{aligned} t_{\text{pdmax}} \text{ from } Q_2 \text{ to } D_1 &: 2\text{ps (XOR)} + 5\text{ps (INV)} = 25\text{ps} \\ \text{from } Q_1 \text{ to } D_2 &: 10\text{ps (AND)} + 10\text{ps (OR)} = 20\text{ps}. \end{aligned}$$

$$\text{thus } t_{\text{setup}} = 5\text{ps} - \underbrace{25\text{ps}}_{t_{\text{pdmax}}} - \underbrace{10\text{ps}}_{t_{\text{pcq}}} = 15\text{ps}$$

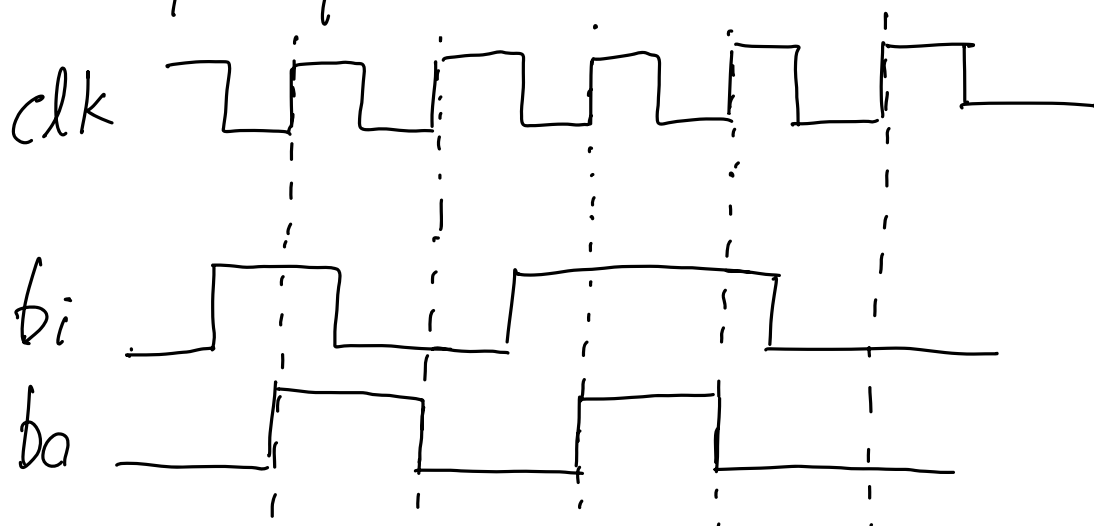
(b) max hold time is given by $t_{\text{pdmin}} + t_{\text{pcq}}$.

$$\begin{aligned} t_{\text{pdmin}} \text{ from } Q_2 \text{ to } D_1 &: 10\text{ps (XOR)} + 2\text{ps (INV)} = 12\text{ps} \\ \text{from } Q_1 \text{ to } D_2 &: 5\text{ps (AND)} + 5\text{ps (OR)} = 10\text{ps}. \end{aligned}$$

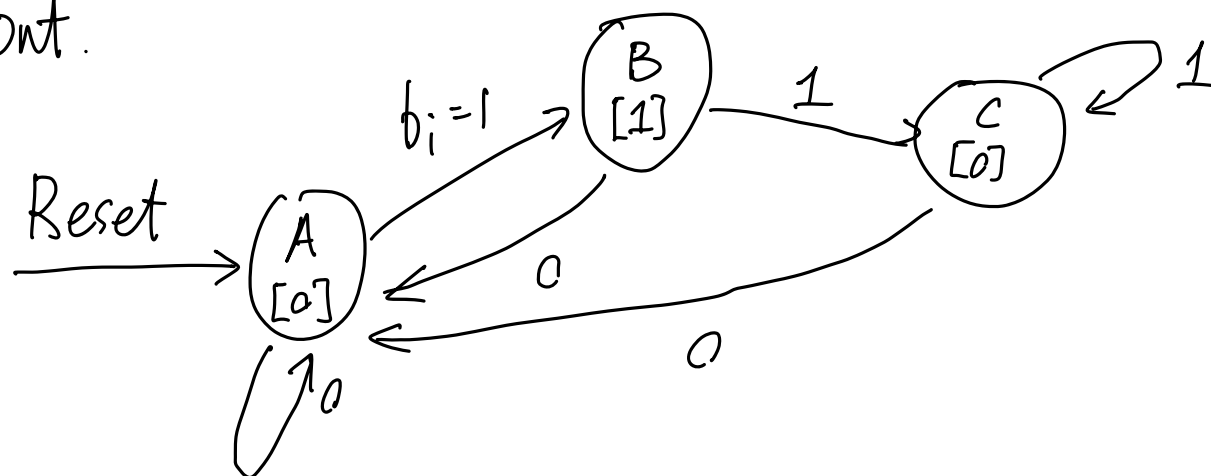
$$\text{thus } t_{\text{hold}} = \underbrace{10\text{ps}}_{t_{\text{pdmin}}} + \underbrace{10\text{ps}}_{t_{\text{pcq}}} = 20\text{ps}.$$

#3

(a) Example sequence:



Taking the button state at each rising edge as input.



state A: idle

state B: triggered / excited by a button click

state C: button held

(b) Transition table:

encode A: 00, B: 01, C: 10

Current state	$b_i = 1$	$b_i = 0$
00	01	00
01	10	00
10	10	00

(c) Table for the next state and output:

Current state	b_i input	next state	b_o output (present)
00	1	01	0
00	0	00	0
01	1	10	1
01	0	00	1
10	1	10	0
10	0	00	0

Set $N_1 N_0$ = the next state code
 $Q_1 Q_0$ = the current state code.

then $N_1 = 1$ when $b_i = 1$ ^{AND} $Q_1 Q_0 = 01$ or 10 .

$N_0 = 1$ when $b_i = 1$ ^{AND} $Q_1 Q_0 = 00$

$$N_1 = b_i (Q_1 + Q_0), \quad N_0 = b_i Q_1' Q_0'$$

$$b_a = Q_1' Q_0$$

(d). Please see the RTL and test bench files submitted (Q3.v and Q3_tb.v). The

Waveform. Notice that the clock time is 1ns

